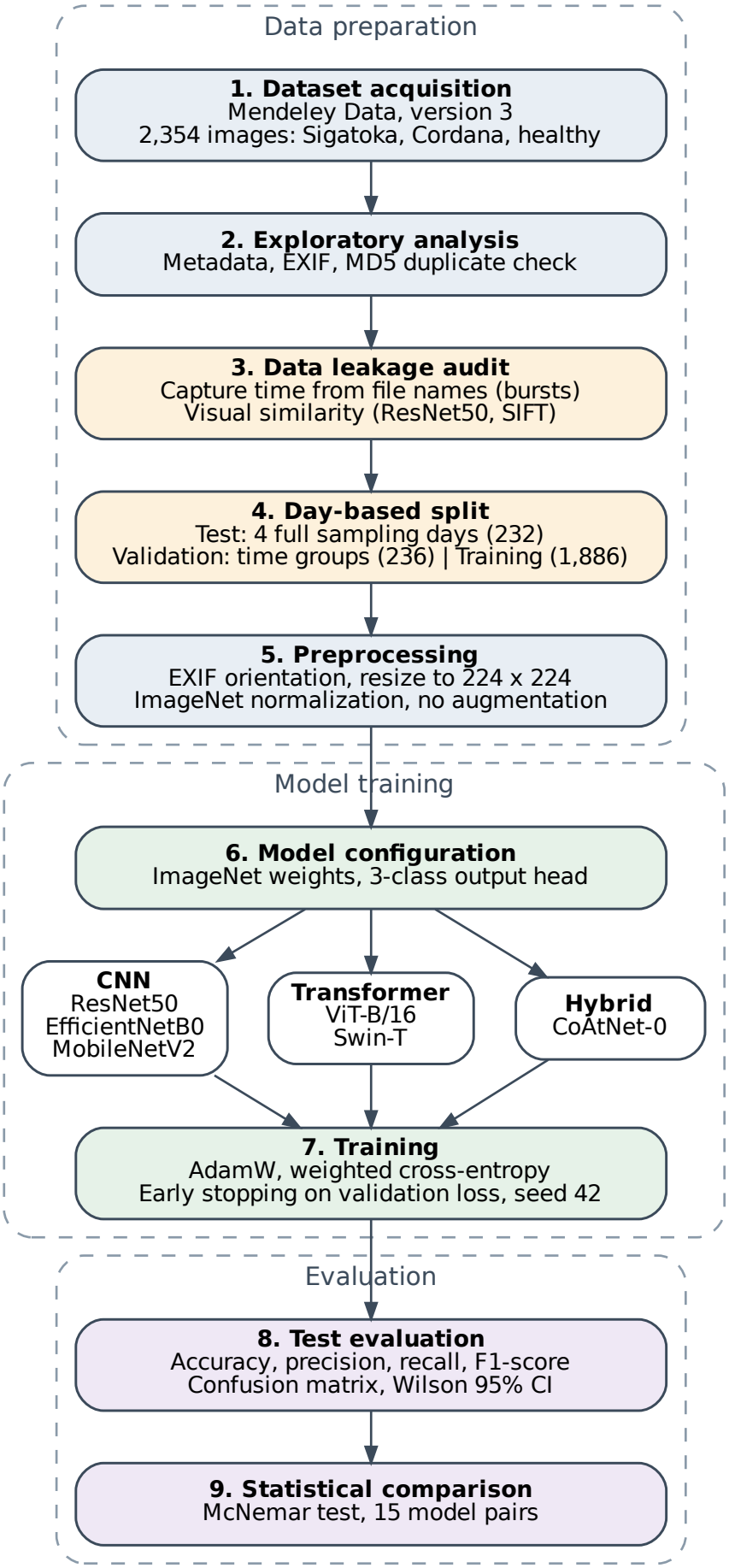
Data preparation
1. Dataset acquisition
Mendeley Data, version 3
2,354 images: Sigatoka, Cordana, healthy
2. Exploratory analysis
Metadata, EXIF, MD5 duplicate check
3. Data leakage audit
Capture time from file names (bursts)
Visual similarity (ResNet50, SIFT)
4. Day-based split
Test: 4 full sampling days (232)
Validation: time groups (236) | Training (1,886)
5. Preprocessing
EXIF orientation, resize to 224 x 224
ImageNet normalization, no augmentation
Model training
6. Model configuration
ImageNet weights, 3-class output head
CNN
ResNet50
EfficientNetB0
MobileNetV2
Transformer
ViT-B/16
Swin-T
Hybrid
CoAtNet-0
7. Training
AdamW, weighted cross-entropy
Early stopping on validation loss, seed 42
Evaluation
8. Test evaluation
Accuracy, precision, recall, F1-score
Confusion matrix, Wilson 95% CI
9. Statistical comparison
McNemar test, 15 model pairs